INCH–POUND

MIL–DTL–51489A(EA)
w/AMENDMENT **3**
8 February 2007
SUPERSEDING
MIL–DTL–51489A(EA)
w/AMENDMENT2
21 November 2006

DETAIL SPECIFICATION

PROJECTILE, 155MM, SMOKE, WP, M110A2, FILLING ASSEMBLY

Inactive for new design after 27 Dec 1995

This specification is approved for use by the U.S. Army Edgewood Chemical Biological Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense within the distribution limitations at the bottom of the page..

1. SCOPE

1.1 Scope. This specification covers the filling assembly with WP, of the 155MM, M110A2 projectile without explosive components.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Edgewood Chemical Biological Center, ATTN: AMSRD–ECB–ENA–S, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010–5424 or emailed to SpecsTeam@apgea.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1320

DISTRIBUTION STATEMENT A. Approved for public release. Distribution unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL−W−215 – White Phosphorus (WP)

STANDARDS

DEPARTMENT OF DEFENSE

MIL−STD−1168 – Ammunition Lot Numbering and Ammunition Data Card
MIL−STD−1916 – DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111−5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY EDGEWOOD CHEMICAL BIOLOGICAL CENTER

DRAWINGS

19203−9217031 – Projectile, 155MM: Smoke, WP, M110A2 Filling Assembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

(Copies of these documents are available from U.S. Army Edgewood Chemical Biological Center, ATTN: AMSRD−ECB−ENA−D, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010−5424.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design and construction.

3.2.1 Materials. All materials cited on Drawing 19203−9217031 and its subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.2.2 Components. All components of the PROJECTILE, 155 MM, SMOKE, WP M110A2, FILLING ASSEMBLY shall conform to the specifications and drawings listed on Drawing 19203−9217031 and subsidiary drawings.

3.2.3 Manufacture and assembly (see 6.5). The PROJECTILE, 155 MM, SMOKE, WP M110A2, FILLING ASSEMBLY shall be manufactured and assembled as shown on Drawing 19203−9217031.

3.3 Filling and assembly.

3.3.1 Filling. The projectile cavity may contain light rust in the form of discoloration or staining with no visual evidence of pitting, etching, flaking or other surface damage, covering not more that 25% of the cavity surface. The projectile, free of volatile corrosion inhibitor paper or any other foreign matter, shall be filled with molten WP conforming to MIL−W−215 (see 6.5) at a temperature of 140˚ ± 10˚F. When water is in contact with the WP during the filling operation, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. The WP shall be solidified with the projectile in a vertical nose−up position.

3.3.2 Filling Height. There shall be a 3 to 5 percent void, of total volume existing above the surface of the water over the WP in the filled and closed projectile. The depth of water remaining in the filled and closed projectile shall not exceed 1/4−inch. The projectile shall be radiographed as specified in 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

3.4 Screening. Oven test will be used to screen for rejects as specificed in 4.4.2.

3.5 Leakage and contamination. Prior to lifting lug installation, the exterior surface of the filled and closed projectile and the interior of the adapter and burster casing shall be cleaned thoroughly to remove all grease, oil, WP and any other contamination. No steel wire brush shall be used on the rotating band (See 6.6). The closed and cleaned projectile shall not leak or show signs of contamination.

3.6 Rotating band. The rotating band shall be free from the following damage or contamination:

(a) There shall be no damage that results in raised brass.

(b) There shall be no paint or foreign matter on the outer diameters.

(c) Nicks, gouges, scratches and other depressions on the lands not exceeding 0.025 inches in depth are acceptable.

(d) Nicks, gouges, scratches and other depressions on the lands with depths greater than 0.025 and less than or equal to 0.050 inches are acceptable unless the damage extends longitudinally across more than half of the land width.

(e) "Washboard" or striated area if it extends longitudinally across more than half of the land width and the depth exceeds 0.025 inches at any point is a reject.

(f) Nicks, gouges, scratches and other depressions on the lands with depths greater than 0.050 are rejects.

3.7 Lot numbering and Ammunition Data Cards. Lot numbering and Ammunition data Cards (see 6.2) shall be prepared in accordance with MIL−STD−1168.

3.8 Workmanship. All parts and assemblies shall be free from contamination (dirt, grease, rust, or foreign matter) and damage (cracks, burrs, chips). The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Care shall be exercised at all times to prevent damage (nicks or burrs) to the projectile rotating band. All required marking and painting shall not be marred or obliterated.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Lot. The first article lot shall consist of 260 filled projectiles or production during one 8−hour shift, whichever is greater. The first article lot shall be manufactured using the same methods, materials, equipment, processes, inspections, in−process tests and facilities as will be used during regular

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

production and shall be submitted for inspection and approval in accordance with the terms of the contract (see 6.3).

4.2.2 Inspections to be performed. The first article sample items shall be examined and tested 100% for all drawing and specification requirements.

4.2.3 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.2.4 Sampling.

4.2.4.1 For examination. Sampling for examination shall be in accordance with Table I.

TABLE I

COMPONENT	DRAWING	SAMPLE SIZE
Projectile, 155 MM; Smoke, WP, M110A2, Filling Assembly	9217031	CRITICAL characteristics−100% All other characteristics−80

4.3 Conformance inspection.

4.3.1 Lotting. Inspection lots shall comply with the provisions of MIL−STD−1916 and MIL−STD−1168, in addition lots of filled projectiles shall contain projectile metal parts assemblies and burster cases of single lot interfix numbers from single manufacturers and WP from one manufacturer.

4.3.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.3.5. Samples shall be selected at random.

4.3.3 Inspection procedure. Sample items shall be examined in accordance with the classification of characteristics in 4.3.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.3.4 Critical characteristics. Unless otherwise specified in the classification of characteristics, the sampling and acceptance criteria for critical characteristics shall be in accordance with the **Critical characteristics** paragraph of MIL−STD−1916. If any critical defects are encountered the **Special reservations for critical nonconformance** paragraph of MIL−STD−1916 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

4.3.5 Classification of characteristics. Conformance examinations shall be as specified in the following classification of characteristics paragraphs. When verification levels (VLs) are specified for selection of sample sizes and acceptance criteria, the attributes sampling plan of MIL−STD−1916 shall apply unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.5(a)	Projectile, 155MM: smoke, WP, M110A2; filling assembly	SHEET 1 OF 1	9217031

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	No WP leakage or contamination	4.3.4	3.5	VI
Major				
101	Depth to burster casing correct	VL IV	3.2	AIE
102	Threads correct	VL IV	3.2	VI/AIE[1]
103	Bourrelet diameter after coating correct (applicable after retouching of paint)	VL IV	3.2	AIE
104	Void correct	VL IV	3.3	4.4.1
105	Water depth correct	VL IV	3.3	4.4.1
106	Marking correct	VL IV	3.2	VI[2]
107	Rotating band free of contamination	VL IV	3.6	VI
108	Ammuntion data card information complete and correct	1; Acc 0, Rej 1	3.7	VI[2]
Minor				
201	Threads greased	VL II	3.2	VI
202	Lifting plug evident, not frozen or loose	VL II	3.2	VI/AIE
203	Protective coating correct	VL II	3.2	VI
204	Workmanship correct	VL II	3.8	VI

NOTES:

AIE − Acceptance inspection equipment

VI − Visual inspection

1. When Fill and LAP contractor is the same, inspection may be checked concurrently with MIL−DTL−60477 Paragraph 4.3.2.4(d) Major 102 “Disc and Cap Assembly Fully Seated on Burster Tube” and Paragraph 4.3.2.4(e) Major 101 “Disassembly Torque of Lifting Plug Assembly“ at the LAP line.

2. When Fill and LAP contractor is the same, may be conducted instead with MIL−DTL−60477 using paragraph 4.3.2.4(e) Minors 203 “Marking evident and correct (except zone markings)” and 204 “Ammunition data card complete and correct” at the LAP line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

4.4 Tests.

4.4.1 Filling height (see 6.4). The filled and closed projectile shall be radiographed to insure proper filling, with the projectile in a vertical, nose−up position and the WP at a temperature of 130˚F ± 10˚F (see 6.5).

4.4.2 Oven screening test (see 6.4). The projectiles shall be placed in an oven in a vertical, nose−up position, stabilized at 212˚ to 217˚F, maintained at the specified temperature 15 minutes, then removed from the oven and immediately observed for leakage. A red to brown residue or the emission of smoke constitutes leakage.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in−house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The projectile, 155MM, WP, M110A2, Filling Assembly is intended for use in the Projectile, 155MM, WP, M110A2.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article:
(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.
(4) When a new or partial first article is required (see 6.3).
(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
(d) Requirements for contractor acceptance inspection equipment (AIE) design ap-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

proval (see 6.4).

(e) Data Item Description, Army Electronic Product Support (AEPS) access information, Governmental Agency Responsible for Acceptance (GARA) information, and distribution for the MIL−STD−1168 Ammunition Data Cards (see 6.5).

(f) Packaging requirements (see drawing 9217031).

(g) Requirements for Ammunition lot numbering (see 3.7).

6.3 Occurrences requiring a new first article inspection. Any change of vendors, methods, materials, equipment, facilities, or component/assembly lot interfix or if there is a 6 month break in production requires a new full or partial first article inspection.

6.4 Submission of inspection procedures and equipment (AIE) designs for approval. Submit inspection procedures and equipment designs, as required, to Director, U.S. Army ECBC, ATTN: AMSRD−ECB−END−S(RI), Rock Island, IL 61229−7390. This address will be specified on the Contractor Data Requirements List, DD Form 1423 in the contract.

6.5 Ammunition Data Card distribution. In addition to the automated distribution; one hard copy with each shipment to each destination, one copy to Director, U.S. Army ECBC, ATTN: AMSRD−ECB−END−S, Aberdeen Proving Ground, MD 21010−5424.

6.6 Caution. Because of the hazardous nature of WP operations, filling installations are advised to take all necessary precautions required by proper industrial hygiene and safety practices to protect personnel from injury while filling projectiles in accordance with this specification.

6.7 Decontamination. Filled and closed projectiles may be decontaminated by passing through a hot water and detergent spray washer, or by flushing with hot water while scrubbing vigorously with steel wool or a wire brush. NOTE, do not use a steel wire brush on the rotating band. Additional cleaning of the burster may be effected with a steam lance and a power driven wire brush.

6.8 Acceptance Inspection Equipment (AIE). Any measuring or test device that is traceable to national or international standards used to assure conformance of material to the contract requirements.

6.9 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

6.10 Standard operation procedures. The Standard Operating Procedures (SOP) or any equivalent document having to do with the management and control of the operating process, shall have a section dedicated to policies and procedures to prevent, control, and dispose of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

substances harmful to the environment, all in accordance with Federal and State environmental laws and regulations.

6.11 Subject term (key word) listing.

Bourrelet
Burster
Plug, lifting
White Phosphorus

Custodian:

Army − EA

Preparing activity:

Army − EA

(Project 1320−2007−004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.